

Introduction to Contract Theory and Auctions

Hidden Information: The basic model of adverse selection

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Adverse Selection

- Uninformed party (principal) offers contract to informed party (agent)
- Examples
 - ▶ Insurance
 - ★ Insuree knows her risk, insurer does not
 - ★ Insurer offers several packages with different premiums and deductibles
 - ▶ Finance
 - ★ Borrower knows the risk of project, lender does not
 - ★ Lender offers several packages with different interest rates and collateral requirements
 - ▶ Hiring
 - ★ Applicants know their ability, employer does not
 - ★ Employer offers different packages varying wages, bonuses, etc.
 - ▶ Pricing
 - ★ Buyer knows her valuation of the product, seller does not
 - ★ Seller offers different qualities at different prices, or quantity discounts

Delegation

- The problem arises in particular when a **principal** *delegates* a task to an **agent**.
- Delegation can be motivated by
 - ▶ The possibility of benefiting from increasing returns associated with the division of tasks.
 - ▶ Principal's lack of ability or time to perform the task himself.
 - ▶ Bounded rationality or complexity issues that the principal faces.
- Because of delegation, the agent may get access to **information** not available to the principal.
 - ▶ The opportunity cost of the task.
 - ▶ The technology.
 - ▶ The ability of the agent to perform the task.
- This information is **private knowledge** of the agent, i.e. there is **adverse selection**

Delegation: Examples

- Both private and public transactions are subject to adverse selection.
 - ▶ Landlord delegates cultivation of his land to a tenant, who will be the only one observing the exact weather conditions.
 - ▶ A client delegates his defense to an attorney, who is the only one that knows the true difficulty of the case.
 - ▶ An investor delegates the management of his portfolio to a broker, who privately knows the prospects of the possible investments.
 - ▶ Stockholders delegate the firm's day-to-day decisions to a manager, who is the only one who observes the precise business conditions.
 - ▶ The Department of Defense procures a good from the military industry without knowing its exact cost structure.
 - ▶ A regulator contracts for service with a Public Utility company without having complete information about its technology

Delegation: common issues

- The information gap between principal and agent has fundamental implications for the design of the bilateral contract that they sign.
- The contract must **elicit** the private information.
- However, this can be done only by giving up some **information rent** to the privately informed agent.
- This information rent distorts the volume of trade achieved under asymmetric information.
- There is an Allocative Efficiency-information rent trade off.
- The first best contract, the one that achieves allocative efficiency is not feasible.
- We will look for second best contracts, the ones that find the optimal efficiency-information rent trade off.

Introduction: the basic model of adverse selection

- One-shot relationship between the principal and the agent.
 - ▶ The parties cannot rely on repetition to achieve efficient trades:
 - ★ Cooperation through repetition.
 - ★ Learning private information through repetition.
 - ★ Reputation effects.
- The short-term relationship implies that the relationship can only be regulated by a **contract**.
- We implicitly assume that there exists a legal framework, a benevolent court of law, which can enforce the contractual relationship.
 - ▶ This feature prevails in most studies of market economies.
 - ▶ If you want to see a model in which only power and coercion regulate the exchange of resources, read Piccione & Rubinstein (2006).

Goals of the basic model of adverse selection

- Characterize the optimal rent extraction-efficiency trade-off of the principal.
- We proceed in two steps:
 - 1 We need a description of the set of allocations that the principal can achieve despite the information gap.
 - ★ An allocation is a specification of output to be produced and a distribution of the gains from trade.
 - ★ The set of achievable allocations can be characterized by **incentive compatibility constraints**.
 - ★ We also impose voluntary participation into the contract: **participation constraints**.
 - ★ Incentive and participation constraints define the set of **incentive feasible allocations**.
 - 2 Analysis of the objective function of the principal under within the set of incentive feasible allocations.
 - ★ In general, incentive constraints are binding which implies that adverse selection impacts the efficiency of trade.

Implicit assumptions

- Several implicit assumptions
 - 1 Both principal and agent are utility maximizers, i.e. rational individualistic agents.
 - 2 The principal doesn't know the agent's private information, but the probability distribution (beliefs) of this information is *common knowledge*
 - ★ The beliefs of the principal regarding the private information are known by the principal and the agent; this fact is known to both; that beliefs are known to both is also a fact that is known to both; the previous fact is also known to both,..., ad infinitum.
 - 3 The principal is a Bayesian expected utility maximizer.
- When designing the contract, the principal moves first as a Stackelberg leader without knowing the private information of the agent.
- The principal needs to anticipate the agent's subsequent behavior depending on his private information and optimize accordingly within the set of available contracts.
- When choosing a contract, the principal **commits** to it.
- The contract is offered at the **interim** stage, i.e. when the agent already knows his private information.

The basic model of adverse selection: formal development

- Technology, preferences and information

- ▶ Suppose a consumer or firm (the principal) wants to delegate the production of q units of a good.
- ▶ The value to the principal is $S(q)$ where $S' > 0$, $S'' < 0$ and $S(0) = 0$.
- ▶ The production cost of the agent is unobservable to the principal, but is common knowledge that the fixed cost is F and that the marginal cost θ is in the set $\Theta = \{\theta_\ell, \theta_h\}$, where $\theta_\ell < \theta_h$.
- ▶ We say that a type θ_ℓ agent is efficient and a type θ_h agent is inefficient, with respective probabilities ψ and $1 - \psi$.
- ▶ The cost function of the agent is:

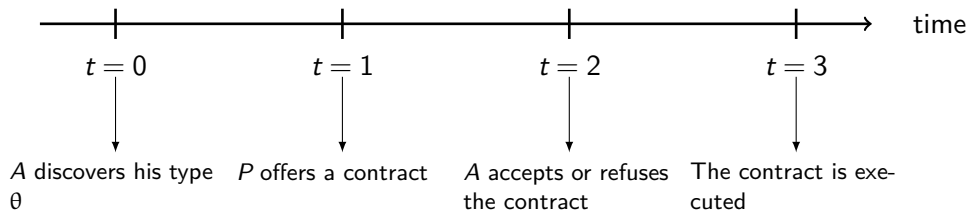
$$C(q, \theta_\ell) = \theta_\ell q + F \quad \text{with probability } \psi$$

$$C(q, \theta_h) = \theta_h q + F \quad \text{with probability } 1 - \psi$$

- ▶ We denote $\Delta\theta = \theta_h - \theta_\ell > 0$ as the spread of uncertainty on the agent's cost.
- ▶ When making his decision, an agent is informed about his type.
- ▶ This information structure is exogenously given to the players.

Contracting variables and timing

- The variables of the problem are q and t , which is a transfer received by the agent.
- Let $\mathcal{A} = \{(q, t) : q \in \mathbb{R}_+, t \in \mathbb{R}\}$ be the set of feasible allocations.
- These variables are observable and verifiable by a third party, so that they can be included in an enforceable contract.
- Timing:



Benchmark: The complete information Optimal contract

- First-Best Production Levels.

- ▶ Define the surplus or social value per type as $W_\ell(q) = S(q) - C(q, \theta_\ell)$, $W_h(q) = S(q) - C(q, \theta_h)$.
- ▶ Recall that $C(q, \theta_i) = \theta_i q + F$, where $\theta_i \in \Theta$.
- ▶ Thus, the efficient levels are obtained by solving the following problems:

$$\max_q W_\ell(q), \quad \text{st.} \quad W_\ell(q) \geq 0$$

$$\max_q W_h(q), \quad \text{st.} \quad W_h(q) \geq 0$$

- ▶ An optimum is obtained from the first-order conditions:

$$S'(q_\ell^*) = \theta_\ell$$

$$S'(q_h^*) = \theta_h$$

- ▶ Optimal social surplus is $W_\ell^* = W_\ell(q_\ell^*)$ and $W_h^* = W_h(q_h^*)$ if these are non-negative.
- ▶ Notice that $W_\ell^* > W_h^*$ (why?)

The optimal contract: features

- The previous result implies that a sufficient condition for the production of the efficient quantities is that

$$W_h^* = S(q_h^*) - \theta_h q_h^* - F \geq 0$$

- Since the fixed cost plays no role in the efficient quantities, except to justify the existence of a single agent, we will set it to zero from now on.
- note that since S' is decreasing, i.e. the marginal value of output to the principal is decreasing, the efficient quantities are such that $q_\ell^* > q_h^*$.

Implementation of the First-best

- We have solved the “Social planner problem” in order to get the efficient quantities.
- The problems that the principal is solving are, for each $\theta_i \in \Theta$

$$V_i^* = \max_{q_i, t_i} S(q_i) - t_i \quad \text{s.t.} \quad t_i - \theta_i q_i \geq 0$$

where (q_i, t_i) is a menu of contracts indexed by type.

- To solve the problem, the principal can offer a **take-it-or-leave-it offer**:
 - ▶ $t_h^* = \theta_h q_h^*$.
 - ▶ $t_\ell^* = \theta_\ell q_\ell^*$.
- Whatever the agent's type, he accepts and makes zero profits.
- The complete information optimal contract is thus (q_h^*, t_h^*) and (q_ℓ^*, t_ℓ^*) .
- Since the principal has all the bargaining power, $V_i^* = W_i^*$ for $i = h, \ell$.

The indifference curves for the agent

- For a fixed level of utility U_i , the indifference curve of type- i agent, where $i = h, \ell$ is defined by

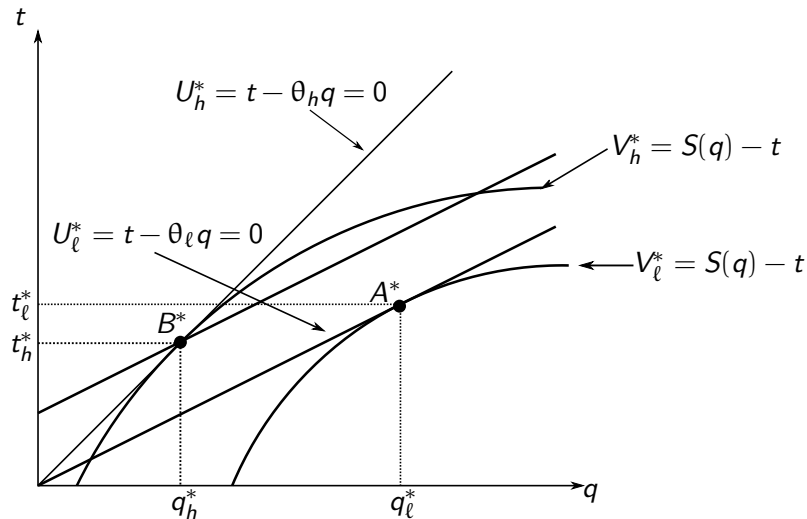
$$U_i = t - \theta_i q$$

- How do the indifference curves look in (q, t) space?
- What is the direction of increase? Northwest
- Note that the slope of the indifference curve is the marginal cost of production of each type, i.e. θ_i .
- Hence, the slope of the inefficient type is larger than for the efficient type.
- Inefficient type has steeper indifference curves compared to the efficient type.
- Therefore, the indifference curves **cross only once**.
- This important property is known as the **single-crossing** or **Spence-Mirrlees** property.

Implementation of the first best

- Note that at the first-best contract:
 - ▶ $t_h^* = \theta_h q_h^*$.
 - ▶ $t_\ell^* = \theta_\ell q_\ell^*$.
- Thus, the utilities of the agents are left at $U_i^* = 0$ at the optimal FB contract.
- Graphically, the isoprofit curves of the principal:
 - ▶ $V_\ell^* = S(q) - t$.
 - ▶ $V_h^* = S(q) - t$.are tangent to the respective indifference curves of the agents.
- As mentioned before, $q_\ell^* > q_h^*$.
- However, the size and ordering of the payments t_ℓ^* and t_h^* depend on the curvature of S .

Graphical Analysis: first-best contract



Incentive Feasible Menu of contracts

- Suppose now that the marginal cost θ is private information.
- Suppose that naively, the principal offers the menu of contracts that we have identified: (t_i^*, q_i^*) for $i = h, \ell$.
- The principal expects for the agents to self-select. What can go wrong?
- Note that the contract (t_h^*, q_h^*) is preferred by both types of agents to the contract (t_ℓ^*, q_ℓ^*) Why?
- Agent's θ_h utility from (t_h^*, q_h^*) is higher than the utility from (t_ℓ^*, q_ℓ^*) .
- Agent's θ_ℓ utility from (t_h^*, q_h^*) is higher than the utility from (t_ℓ^*, q_ℓ^*) .
- Offering the menu (t_i^*, q_i^*) for $i = h, \ell$. fails to have the agents self-select: both choose (t_h^*, q_h^*) .
- We say that the contract (t_i^*, q_i^*) for $i = h, \ell$ is **not incentive compatible**.

Incentive compatibility and participation

Definition 1

A menu of contracts $\{(t_\ell, q_\ell), (t_h, q_h)\}$ is **incentive compatible** when (t_ℓ, q_ℓ) is weakly preferred to (t_h, q_h) by agent θ_ℓ and (t_h, q_h) is weakly preferred to (t_ℓ, q_ℓ) by agent θ_h .

Formally, this definition amounts to the fact that allocations must satisfy the following **incentive compatibility constraints**:

$$t_\ell - \theta_\ell q_\ell \geq t_h - \theta_\ell q_h$$

$$t_h - \theta_h q_h \geq t_\ell - \theta_h q_\ell$$

Definition 2

For the agents to participate, i.e. accept the menu, the menu must yield to each agent at least his outside opportunity level. The following **participation constraints** must be satisfied:

$$t_\ell - \theta_\ell q_\ell \geq 0$$

$$t_h - \theta_h q_h \geq 0$$

Special classes of contracts

Pooling or Bunching Contracts

Suppose that the contracts targeted for each type coincide:

$$t_h = t_\ell = t_P, \quad q_h = q_\ell = q_P$$

The incentive constraints are all trivially satisfied by these contracts, but at the cost of flexibility in allocations. Only participation contracts matter for this type of contract. Note that the participation constraint for the inefficient type implies the one for the efficient type (why?)

Shutdown of the Least Efficient type

Suppose that

$$t_h = q_h = 0, \quad t_\ell = t_s, q_\ell = q_s$$

This contract implies **shutdown of the least efficient type**. This contract also reduces the number of constraints but at the cost of excessive screening of the types.

Monotonicity Constraints

- Incentive compatible quantities must generally satisfy a **monotonicity constraint** in well behaved problems.
- Adding up the incentive compatibility constraints yields

$$q_\ell \geq q_h$$

- **Remark: this result is independent of the principal's preferences!**
- The monotonicity constraint is usually called an **implementability condition**
- If a pair (q_ℓ, q_h) is implementable then it must satisfy the monotonicity condition. For our simple example this condition is also sufficient.
- Sufficiency: suppose monotonicity holds, then it is clear there exist transfers t_h and t_ℓ such that the incentive compatibility constraints hold: just take transfers such that

$$\theta_h(q_h - q_\ell) \leq t_h - t_\ell \leq \theta_\ell(q_h - q_\ell)$$

Information Rents

- We have seen that at the First-Best contract, the utility levels of the agents satisfy $U_i^* = t_i^* - \theta_i q_i^* = 0$ for $i = h, \ell$.
- This is in general not possible anymore under incomplete information.
- Take any menu $\{(t_h, q_h), (t_\ell, q_\ell)\}$ and consider the utility level that a type θ_ℓ agent would get by mimicking a θ_h agent. He would get:

$$t_h - \theta_\ell q_h = t_h - \theta_h q_h + (\Delta\theta)q_h = U_h + (\Delta\theta)q_h$$

- Note that even if $U_h = 0$, the efficient type benefits from an information rent $(\Delta\theta)q_h$ coming from his ability to mimic the inefficient type.
- The goal is to determine the optimal information rent to leave to the efficient type.
- Notation (information rents):
 - ▶ $U_\ell = t_\ell - \theta_\ell q_\ell$.
 - ▶ $U_h = t_h - \theta_h q_h$.

The optimization problem of the principal

- The principal will offer a menu of contrants $\{(q_\ell, t_\ell); (q_h, t_h)\}$ in $\mathcal{A}$ that solve the following optimization problem:

$$\begin{aligned} \max_{\{(q_\ell, t_\ell); (q_h, t_h)\}} \quad & \psi(S(q_\ell) - t_\ell) + (1 - \psi)(S(q_h) - t_h) \\ \text{subject to :} \quad & t_\ell - \theta_\ell q_\ell \geq t_h - \theta_\ell q_h \\ & t_h - \theta_h q_h \geq t_\ell - \theta_h q_\ell \\ & t_\ell - \theta_\ell q_\ell \geq 0 \\ & t_h - \theta_h q_h \geq 0 \end{aligned}$$

- Note the implicit restrictions: the menu $\{(q_\ell, t_\ell); (q_h, t_h)\}$ needs to belong to $\mathcal{A}$, thus quantities are non-negative.

Using rents instead of transfers

- It will be instructive to rewrite the problem of the principal using information rents instead of transfers.
- Using this transformation allows us to see more clearly the trade-off that the principal is facing.
- Recall:
 - ▶ $U_\ell = t_\ell - \theta_\ell q_\ell \rightarrow t_\ell = U_\ell + \theta_\ell q_\ell.$
 - ▶ $U_h = t_h - \theta_h q_h \rightarrow t_h = U_h + \theta_h q_h.$
- The principal will now choose a menu which is composed of a **distribution of the gains from trade** between the principal and the agent and a quantity: $\{(q_\ell, U_\ell); (q_h, U_h)\}$
- Rewriting the objective function of the principal in terms of quantities and gains from trade yields:

$$\begin{aligned} & \psi(S(q_\ell) - U_\ell - \theta_\ell q_\ell) + (1 - \psi)(S(q_h) - U_h - \theta_h q_h) = \\ & \underbrace{\psi(S(q_\ell) - \theta_\ell q_\ell) + (1 - \psi)(S(q_h) - \theta_h q_h)}_{\text{Expected Allocative Efficiency}} - \underbrace{(\psi U_\ell + (1 - \psi) U_h)}_{\text{Expected Information Rent}} \end{aligned}$$

Rewriting the constraints using information rents & the optimization program

- We now proceed to rewrite the incentive compatibility and participation constraints in terms of the variables U_ℓ and U_h .
- Incentive compatibility constraints:
 - ▶ $U_\ell \geq U_h + \Delta(\theta)q_h$.
 - ▶ $U_h \geq U_\ell - \Delta(\theta)q_\ell$.
- Participation constraints:
 - ▶ $U_\ell \geq 0$.
 - ▶ $U_h \geq 0$.
- Thus, the optimization program (P) using the rewritten objective and constraints is:

$$\max_{\{(q_\ell, U_\ell); (q_h, U_h)\}} \psi(S(q_\ell) - \theta_\ell q_\ell) + (1 - \psi)(S(q_h) - \theta_h q_h) - (\psi U_\ell + (1 - \psi) U_h)$$

$$\text{subject to : } U_\ell \geq U_h + \Delta(\theta)q_h$$

$$U_h \geq U_\ell - \Delta(\theta)q_\ell$$

$$U_\ell \geq 0$$

$$U_h \geq 0$$

We will index the solutions for program (P) with the superscript SB (second best).

Simplifying the optimization program

- The main challenge is to determine which constraints are the relevant ones, i.e. which are the binding constraints.
- One approach: check that the program is concave and use the Lagrangian together with the Kuhn-Tucker conditions to solve the problem.
- We will follow a more pragmatic route: we will guess (formally) which constraints will be binding and then will check **ex post** that the omitted constraints are strictly satisfied.
- This route also provides more intuition about the structure of the problem.

Simplifying the optimization program

- Consider firsts contracts without shutdown, i.e. $q_h > 0$.
- Firstly: note that constraints $U_\ell \geq U_h + \Delta\theta q_h$ and $U_h \geq 0$ imply constraint $U_\ell \geq 0$.
 - ▶ This is because the efficient type θ_ℓ can always mimic the inefficient type θ_h .
 - ▶ If a menu of contracts enables the inefficient type to reach his status quo utility level, the efficient agent can reach a strictly higher utility level since he can produce at lower cost.
- The incentive constraint for the inefficient type $U_h \geq U_\ell - \Delta\theta q_\ell$ will also be satisfied.
 - ▶ From the graphical analysis of the First-Best, this one is irrelevant.
 - ▶ The difficulty in the problem comes from an efficient type willing to claim he is inefficient, rather than the other way around.

Simplifying the optimization program

- We will claim now that the two remaining constraints: $U_\ell \geq U_h + \Delta\theta q_h$ and $U_h \geq 0$ hold with equality.
 - ▶ $U_\ell = \Delta\theta q_h$.
 - ▶ $U_h = 0$.
- Towards a contradiction, suppose not, i.e. $\exists \epsilon > 0$ such that:
 - ▶ $U_h = \epsilon$
 - ▶ $U_\ell = \Delta\theta q_h + \epsilon$
- We analyze each case in turn:
 - ▶ Then, the principal can decrease U_h and consequently also decrease U_ℓ (this last statement comes from the other incentive constraint $U_\ell \geq U_h + \Delta\theta q_h$). The gain from this is ϵ , thus $U_h = 0$.
 - ▶ Now suppose $U_\ell = \Delta\theta q_h + \epsilon$, then the principal can decrease U_ℓ by ϵ and gain $\psi\epsilon$. Thus, $U_\ell = \Delta\theta q_h$ is optimal.

Simplifying the optimization program

- Substituting the binding constraints and ignoring the other constraints that we guessed hold strictly yields the following **reduced program** (P')

$$\max_{(q_\ell, q_h)} \psi(S(q_\ell) - \theta_\ell q_\ell) + (1 - \psi)(S(q_h) - \theta_h q_h) - \psi \Delta \theta q_h.$$

- The asymmetry of information causes the principal to forfeit some rent to the efficient type.
- This rent depends on the level of production asked from the inefficient type.
- The inefficient type gets no rent, but the efficient one gets the rent that he could obtain by mimicking the inefficient type.

Solution to the reduced program

$$\max_{(q_\ell, q_h)} \psi(S(q_\ell) - \theta_\ell q_\ell) + (1 - \psi)(S(q_h) - \theta_h q_h) - \psi\Delta\theta q_h.$$

- First order conditions
 - ▶ For q_ℓ : $S'(q_\ell^{\text{SB}}) = \theta_\ell$ or $q_\ell^{\text{SB}} = q_\ell^*$.
 - ▶ For q_h : $(1 - \psi)(S'(q_h^{\text{SB}}) - \theta_h) = \psi\Delta\theta$.
- The first order condition with respect to q_h represents the trade-off that the principal faces between efficiency and rent extraction.
- The expected marginal efficiency gain needs to equal the expected marginal cost brought about by an increase of the inefficient type's output are equated.
- Notice that the omitted incentive constraint of an inefficient agent is satisfied: $0 \geq \Delta\theta q_h^{\text{SB}} - \Delta\theta q_\ell^{\text{SB}}$, which follows from monotonicity of the second best outputs:

$$q_\ell^{\text{SB}} = q_\ell^* > q_h^* > q_h^{\text{SB}}.$$

Summary of the results

Properties of the second-best contract

Under asymmetric information, the optimal contract:

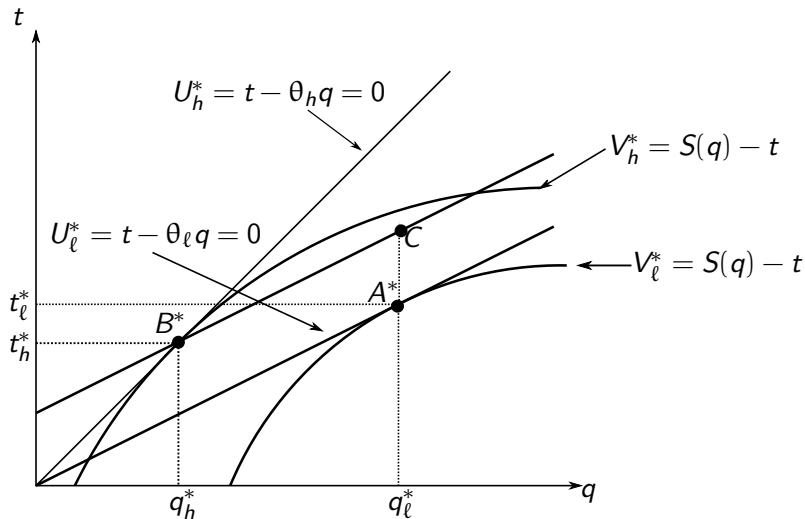
- Has no output distortion for the efficient type with respect to the first best: $q_\ell^{\text{SB}} = q_\ell^*$ and has a downward distortion for the inefficient type, $q_h^{\text{SB}} < q_h^*$ with:

$$S'(q_h^{\text{SB}}) = \theta_h + \frac{\psi}{1 - \psi} \Delta\theta$$

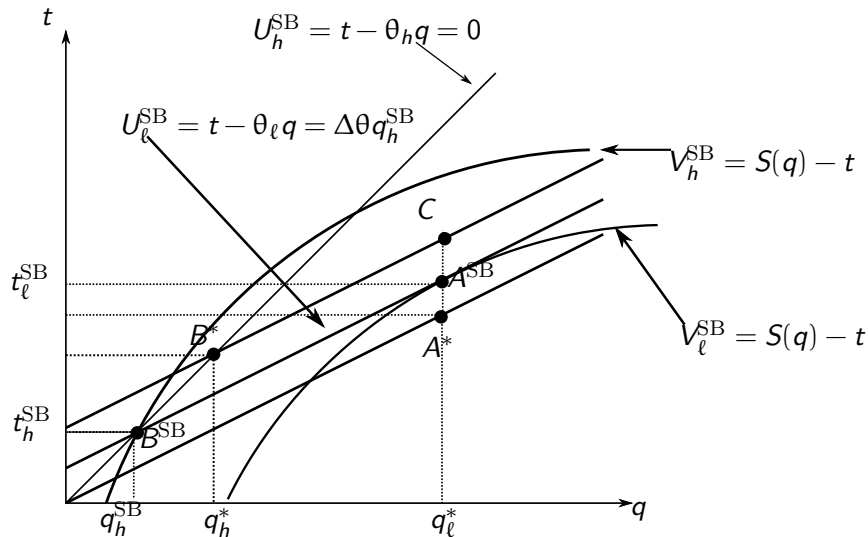
- The efficient type is the only one that gets a positive information rent: $U_\ell^{\text{SB}} = \Delta\theta q_h^{\text{SB}}$.
- The second-best transfers are

$$t_\ell^{\text{SB}} = \theta_\ell q_\ell^* + \Delta\theta q_h^{\text{SB}}, \quad t_h^{\text{SB}} = \theta_h q_h^{\text{SB}}.$$

Graphical Analysis: second-best contract



Graphical Analysis: second-best contract



Shutdown Policy

- If the equation

$$(1 - \psi)(S'(q_h^{SB}) - \theta_h) = \psi\Delta\theta,$$

has no positive solution, q_h^{SB} should be set at zero.

- Notice that in this case, points A_h^{SB} and A^* coincide in the last figure.
- NO rent is given up to the θ_ℓ firm at the contract (q_ℓ^*, t_ℓ^*) selected by this agent.
- Shutdown occurs for the inefficient type θ_h .
- In general, we have two considerations:
 - ▶ More inefficiency emerges because type θ_h does not produce.
 - ▶ On the other hand, there is a gain for the principal since no rent is given for the efficient type θ_ℓ .
- In general, when is shutdown optimal?

Shutdown policy

- For the Shutdown policy:
 - ▶ Type θ_ℓ : (q_ℓ^*, t_ℓ^*)
 - ▶ Type θ_h : $(0, 0)$.
- Thus, shutdown is optimal when:

$$\psi(S(q_\ell^*) - \theta_\ell q_\ell^*) \geq \psi(S(q_\ell^{\text{SB}}) - \theta_\ell q_\ell^{\text{SB}} - \Delta\theta q_h^{\text{SB}}) + (1 - \psi)(S(q_h^{\text{SB}}) - \theta_h q_h^{\text{SB}})$$

- Remember that $q_\ell^* = q_\ell^{\text{SB}}$ so the expression above simplifies to:

$$\psi\Delta\theta q_h^{\text{SB}} \geq (1 - \psi)(S(q_h^{\text{SB}}) - \theta_h q_h^{\text{SB}})$$

- LHS: expected cost of the efficient type's rent due to contracting with the inefficient type when the latter produces a positive amount.
- RHS: expected benefit from transacting with the inefficient type at the second-best level of output.

Asymmetric Information and Marginal Cost Pricing

- We can reinterpret the problem as thinking the principal as a "consumer" and the agent as a firm producing a consumption good.
- The first-best rules can be understood as **price equals marginal cost** since the consumer will equate marginal utility of consumption to price.
- Under information asymmetry, price equals marginal cost only when the firm is efficient ($\theta = \theta_\ell$).
- We see that there is a distortion for the inefficient firm:

$$p(\theta_h) = \theta_h + \frac{\psi}{1-\psi} \Delta\theta$$

- Price is higher than marginal cost in order to decrease q_h produced by the inefficient firm in order to reduce the efficient firm's information rent.
- We can say that price is equal to **a generalized or virtual marginal cost** which includes, in addition to the marginal cost θ_h of the inefficient type, an information cost that is worth: $\frac{\psi}{1-\psi} \Delta\theta$.
- This generalizes the notion of cost to include the **information cost** imposed by asymmetric information.

Are complex contracts better?

- Notice that in the analysis, we restricted the search for an optimum to a menu of contracts, one for each possible type.
- Could a better outcome could be achieved with a much more complex contract?
- Contracts offering the agent to choose among more options?
- What about introducing communication possibilities between the agent and the principal to transmit information, so that the principal can later recommend outputs and payments as a function of transmitted information?
- We will no introduce the notion of a general mechanism allowing for this possibilities.